Spring problem (rubber)

$$f = k\,l$$

force (tension) — deformation

with $k =$ ~~constant~~ independent of l.

$$dE = TdS + Jdx = TdS + f\,dl$$

$$\Rightarrow dF = -S\,dT + f\,dl$$

Find C_l (heat capacity at fixed l) and C_f (at fixed tension f)

<u>At $T = \text{const}$</u> $\quad dF = f\,dl = k\,l\,dl$

$$\Rightarrow F_l(T) = F_0(T) + \frac{1}{2}kl^2$$

free energy at l $\quad (l=0)$

$$\text{Entropy } S = -\left(\frac{\partial F}{\partial T}\right)_l = -\underbrace{\left(\frac{\partial F_0}{\partial T}\right)_l}_{S_0} - \left(\frac{\partial}{\partial T}\frac{1}{2}kl^2\right)_l =$$

$$\Rightarrow S(T) = S_0(T) - \frac{l^2}{2}\left(\frac{\partial k}{\partial T}\right)_l$$

$$\Rightarrow C_l = T\left(\frac{\partial S}{\partial T}\right)_l = T\left(\frac{\partial S_0}{\partial T}\right)_l - T\left(\frac{\partial}{\partial T}\frac{l^2}{2}\left(\frac{\partial k}{\partial T}\right)_l\right)_l =$$

$$= C_0 - \frac{l^2 T}{2}\frac{d^2k}{dT^2}$$

Then $C_f = T\left(\frac{\partial S}{\partial T}\right)_f \Rightarrow$ need $S\big|_f = S_0 - \frac{f^2}{2k^2}\frac{dk}{dT}$

$$l = \frac{f}{k}$$

$$\Rightarrow C_f = T\left(\frac{\partial S_0}{\partial T}\right)_f - T\left(\frac{\partial}{\partial T}\frac{f^2}{2k^2}\frac{dk}{dT}\right)_f =$$

$$= C_0 - \frac{Tf^2}{2}\underbrace{\frac{d}{dT}\left(\frac{1}{k^2}\frac{dk}{dT}\right)}_{\frac{1}{k^2}\frac{d^2k}{dT^2} - \frac{2}{k^3}\left(\frac{dk}{dT}\right)^2}$$

$$C_f = C_0 - \frac{Tf^2}{2k^2}\frac{d^2k}{dT^2} + \underbrace{\frac{Tf^2}{k^3}}_{\frac{T\,k^2\ell^2}{k^3} = \frac{T\ell^2}{k}}\left(\frac{dk}{dT}\right)^2$$

$$\Rightarrow C_f - C_\ell = \frac{T\ell^2}{k}\left(\frac{dk}{dT}\right)^2 > 0$$

Internal energy: $E = F + TS = F_0 + \frac{1}{2}k\ell^2 + \cancel{TS_0}$

$$+T\left(S_0 - \frac{\ell^2}{2}\frac{dk}{dT}\right) = \underbrace{F_0 + TS_0}_{E_0(T)} +$$

$$+\frac{1}{2}k\ell^2\left(1 - \frac{T}{k}\frac{dk}{dT}\right)$$

Thus energy changes during isothermal process is not equal to the work done!

Can be bigger or smaller, depending on $\frac{dk}{dT}$.

$\Rightarrow$ $đQ \neq 0$ for $T = \text{const}$ processes!

Model: For the rubber $f = k\ell = \frac{k_B T}{N d^2}\ell = \underbrace{\tilde{k} T}_{k}\ell$

$$\Rightarrow \frac{dk}{dT} = \tilde{k} \quad , \quad \frac{T}{k}\frac{dk}{dT} = \frac{T}{k}\tilde{k} = \frac{k}{k} = 1$$

$$\Rightarrow E(T) = E_0(T) + \frac{1}{2}k\ell^2(1-1) = E_0(T) \Rightarrow \Delta E = 0$$

due to work $\Delta W = \frac{1}{2}k\ell^2$.

$$dE = đQ + đW$$

$$\Delta Q = \Delta E - \Delta W = -\Delta W < 0 \,!$$

heat is absorbed from outside

for $f = \text{const} = \tilde{k}\, T\, \ell$: increase $T \Rightarrow \ell$ decrease
descrease $T \Rightarrow \ell$ increase

$\Rightarrow$ increasing ℓ results in absorption of heat ΔQ in order to keep the temperature constant!